

Operating System Security Fundamentals (Linux & Windows)

Operating System (OS) security focuses on protecting the system from unauthorized access, malware, and misuse. Securing the operating system is important because it controls hardware, software, user access, and system resources.

User Accounts and Privileges

Operating systems provide different types of user accounts to control access.

Administrator User

- Has full control over the system
- Can install software and change system settings

Standard User

- Has limited access
- Cannot make major system changes

Importance:

Using standard users for daily work improves security and prevents accidental system damage.

File Permissions

File permissions define who can access files and what actions they can perform.

Linux File Permissions

- **Read (r):** View file content
- **Write (w):** Modify file
- **Execute (x):** Run file

Common Linux Commands

- `ls -l` → View file permissions
- `chmod` → Change file permissions
- `chown` → Change file ownership

Purpose:

File permissions prevent unauthorized access to sensitive files.

Administrator vs Standard User

Feature	Administrator	Standard User
System access	Full	Limited
Install software	Yes	No
Change settings	Yes	No

Using the **least privilege principle** increases security.

Firewall

A firewall monitors and controls incoming and outgoing network traffic.

Windows Firewall

- Windows Defender Firewall protects the system
- Blocks unauthorized network connections

Benefit:

Firewalls prevent hackers from accessing the system remotely.

Running Processes and Services

Processes are programs currently running on the system.

Windows Task Manager

- Shows running applications and background services
- Helps identify suspicious processes

Monitoring processes helps detect malware or unnecessary applications.

Disabling Unnecessary Services

Unnecessary services increase the attack surface.

Examples:

- Unused startup programs
- Background services not required by the user

Advantage:

Disabling them improves performance and reduces security risks.

OS Hardening Best Practices

OS hardening means strengthening system security.

Best Practices

- Use strong passwords
- Enable firewall
- Keep OS updated
- Use antivirus software
- Disable unused services
- Follow least privilege principle